Lesson 1.1: AI-powered analytics with Microsoft Copilot

What you learn

- **Copilot in Excel.** Ask questions in natural language to summarise data, suggest charts and spot trends; availability depends on your Microsoft plan.
- **Prompt used.** 'Summarize quarterly growth trends and highlight the top 5 outlier product categories by margin percentage.'
- **Verify.** Check any AI result against the data; the code below shows how to find outliers yourself with the IQR rule.

Tools and resources (official sites, not checked live)

- Copilot in Excel

ASSIGNMENT

Practice for lesson 1.1

Business problem: Find margin outliers two ways.

Steps

1. Run the prompt in Copilot if available.
2. Run the tested code.
3. Compare the answers.

Expected output: The same outlier product from both methods.

SOLUTION (ANSWER KEY)

PYTHON

```
import pandas as pd
df = pd.DataFrame({"product":["A","B","C","D","E","F"],"revenue":
[100,120,90,110,95,900],"cost":[60,70,50,65,55,600]})
df["margin_pct"] = ((df["revenue"]-df["cost"])/df["revenue"]*100).round(1)
q1, q3 = df["margin_pct"].quantile([0.25, 0.75]); iqr = q3-q1
outliers = df[(df["margin_pct"] < q1-1.5*iqr) | (df["margin_pct"] > q3+1.5*iqr)]
print(df.describe().loc[["mean","min","max"]]); print("outliers:",
outliers["product"].tolist())
assert df["margin_pct"].iloc[0] == 40.0
```

Product F has a 33.3 percent margin while the others are near 40 percent, so the rule flags it.

Check yourself

1. Why verify AI analysis?
   Answer: It can be wrong.
2. What is the IQR rule?
   Answer: Outliers lie beyond 1.5 times the interquartile range from the quartiles.

Lesson 1.1 of the **Data Analytics and Data Science** lessons. See this lesson and its official links on skillnavora.in · All chapters and lessons · Open the full lesson guide. Tools, prices and plans change; confirm on each official site.